D5 – Optimization and Ethics Report

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1. Performance Optimization

1.1 Model Performance

The final model is a **calibrated Logistic Regression** classifier. It was selected after comparing 9 models under identical folds and preprocessing.

Metric	Value
PR-AUC	0.7803
ROC-AUC	0.7191
Brier Score	0.2090
Recall (threshold 0.5)	0.7133
Precision (threshold 0.5)	0.7202

Comparison against baselines:

Model	PR-AUC
Dummy (most frequent)	0.5809
Business Rule (recency < 30)	0.6307
Our model	**0.7803**

Our model improves PR-AUC by **+0.15** over the business rule baseline.

1.2 Hyperparameter Tuning

We used **Optuna** with Bayesian search (TPE sampler) and median pruning. 30 trials were run per model. All trials were logged to MLflow.

Best parameters for Logistic Regression:

- `C = 0.0187`
- `solver = lbfgs`
- `max_iter = 190`

1.3 Calibration and Threshold Selection

We calibrated the model using **Platt scaling** (sigmoid) to make the probabilities more reliable.

The optimal decision threshold was selected by maximizing expected profit:

Threshold	Expected Profit
0.5 (default)	$72,032
0.01 (optimal)	**$100,232**
Improvement	**+$28,200 (+39%)**

Cost assumptions:

- Coupon cost: $4
- Retained customer value: $180

1.4 Latency Optimization

Endpoint	p50 Latency	p95 Latency
`/status`	~10 ms	~30 ms
`/predict`	~50 ms	~150 ms
`/predict/batch` (10 items)	~200 ms	~500 ms

The API loads the model **once at startup** using the FastAPI lifespan pattern, so there is no per-request model loading overhead.

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2. Scalability

2.1 Current Capacity

Aspect	Value
Model size	~5 MB
Memory usage	~200 MB
Concurrent requests	Tested up to 10/sec
Cold start	~50 sec (free tier sleep)

2.2 Breaking Points

Load	Expected Behaviour
< 100 req/sec	Works on current setup (1 CPU, 512 MB)
100–1000 req/sec	Requires scaling up (more CPU/RAM)
> 1000 req/sec	Requires horizontal scaling + load balancer

2.3 Cost per 1,000 Requests

Platform	Estimated Cost per 1k Requests
Render (free tier)	$0 (within limits)
AWS App Runner	~$0.02
AWS SageMaker Endpoint	~$0.10

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3. Interpretability

3.1 Feature Importance

The top 5 features (by absolute coefficient magnitude in the Logistic Regression):

Rank	Feature	Direction
1	`days_since_last_purchase`	Negative (more days → less likely to return)
2	`order_count`	Positive (more orders → more likely)
3	`total_spend`	Positive
4	`unique_products_bought`	Positive
5	`avg_days_between_orders`	Negative

3.2 Per-Prediction Explanation

The model provides a probability (0–1) for each customer. This is used to rank customers for the marketing team. The human reviewer can see:

- The probability
- The decision (Send coupon / No coupon)
- The optimal threshold (0.01)

3.3 Is This Fit to Show a Customer?

No – the raw model output is not shown to customers. It is used internally by the marketing team, with a human review step before any coupon is sent.

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4. Fairness Audit

See `GOVERNANCE_LOI25.md` for the full audit. Summary:

Subgroup	Recall Gap	Severity
Spend Q1 (lowest)	-37.3%	🔴 Critical
Spend Q2	-21.0%	🔴 Critical
Tenure 0–6 months	-12.7%	🟡 Moderate

Candidate mitigations:

1. **Lower the threshold** for flagged groups (immediate, no data collection needed).
2. **Collect more data** for new and low-spend customers (3–6 months).
3. **Add features** capturing low-spend and new-customer behaviour.
4. **Separate model** for new customers.

Recommendation: Start with mitigation #1 (lower threshold).

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5. Law 25 Governance

See `GOVERNANCE_LOI25.md` for full details. Summary:

Requirement	Status
Automated-decision register	✅ Provided
Plain-language customer notice (EN/FR)	✅ Provided
Human review step	✅ Marketing manager approves final list
Data minimisation audit	✅ All 25 features justified
Fairness audit	✅ Complete
Model card	✅ `MODEL_CARD.md`
Data card	✅ `DATA_CARD.md`

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6. Ethics Summary

Aspect	Assessment
Human oversight	✅ Present
Data minimisation	✅ Applied
Fairness gaps identified	✅ Yes
Gaps mitigated	⏳ Partially (mitigations proposed)
Transparency	✅ Documents provided
Customer rights	✅ Documented (bilingual)
Compliance with Law 25	✅ Yes

Honest conclusion: The model is fit for deployment **with human review**. Fairness gaps exist but are documented with clear mitigation paths. We do not hide the gaps – we report them and propose solutions.

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7. Summary

Requirement	Status
Performance optimization	✅ Documented
Scalability analysis	✅ Documented
Interpretability	✅ Documented
Fairness audit	✅ Documented
Law 25 governance	✅ Documented
Model and data cards	✅ Provided

End of D5 Report